Pedagogical Support Material

Quantum Mechanics & Quantum Chemistry Programme

For Students and Instructors

Section 1: Learning Tiers (HS → PhD) — Definitions, Competencies and a Worked Example from Bra-Ket Module I.2

Section 2: Assessment Artifact Set — Full Taxonomy, Templates, Rubrics, and Packaging Standard

This document is a companion to the Course Syllabus.
It sets out the shared language for how *material is taught and assessed across all Series and Modules of the programme.*

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Introduction

This document serves two audiences simultaneously.

For students: It explains exactly what is expected of you at each stage of your education, and what kinds of tasks you will be asked to perform. Understanding the tier system and the artifact taxonomy before a course begins removes ambiguity about difficulty levels and grading expectations.

For instructors: It provides a reusable, course-independent specification for how to differentiate content and assessment. The tier system and artifact set described here apply uniformly across *every lecture* in *every module* of the programme. Consistency of language across Series I, II, and III is a deliberate design choice.

How the two sections connect. Section 1 defines the *audience* dimension: who is in the room and what they can be expected to bring. Section 2 defines the *assessment* dimension: what tasks reveal mastery at each level. Together they answer the two questions every educator and student needs:

- *What level is this material pitched at?*
- *What does "mastery" look like for me at my level?*

1 Learning Tiers

Overview. The programme is designed to be taught to a *mixed cohort*: a single lecture or module may be attended simultaneously by high-school students with strong mathematical aptitude and PhD researchers extending their foundations. The five-tier system ensures every student finds appropriate entry points, stretch goals, and assessment tasks, without fragmenting the cohort into separate tracks.

The tiers are *not grade labels*. They describe the prerequisite competency set a student brings and therefore what kind of engagement they can immediately sustain. A student may operate at different tiers for different topics (e.g. Tier 3 for linear algebra, Tier 2 for group theory).

1.1 Tier Ladder at a Glance

Tier	Label	One-line description
1	High School (HS)	Amplitudes are complex arrows; probabilities are squared lengths
2	Beginning UG (BegUG)	Bases and components; compute probabilities with $\langle n\|\psi\rangle$
3	Advanced UG (AdvUG)	Inequalities; geometry and internal consistency of the formalism
4	Master's (MSc)	Rays and projective space; symmetries via Wigner's theorem
5	PhD	Rigorous Hilbert space; separability; domain subtleties

The five tiers form a *cumulative* ladder: every competency at Tier k is assumed at Tier $k+1$. Instructors should always state which tier(s) a given lecture segment targets.

1.2 Tier Definitions and Competencies

Tier 1 — High School (HS)

Goal: Build the *intuition tools* needed for amplitudes and interference.

Core competencies:

- Complex numbers as "2D arrows": $a + ib$, magnitude $|z|$, phase $\arg z$
- **Magnitude squared** as probability: $P \propto |A|^2$
- **Vector addition analogy**: adding complex amplitudes before squaring gives interference
- Basic probability: events, normalisation ("total probability = 1")

You should be able to:

- Compute $|a + ib|^2$
- Understand why $|A_1 + A_2|^2 \neq |A_1|^2 + |A_2|^2$ in general

Tier 2 — Beginning Undergraduate (BegUG)

Goal: Become fluent in the *working language* of bras, kets, bases, and probabilities.

Core competencies:

- Normalisation: $\langle\psi|\psi\rangle = 1$
- Orthonormal bases: $\langle m|n\rangle = \delta_{mn}$

- Components as inner products: $c_n = \langle n|\psi\rangle$
- Probabilities: $P(n) = |c_n|^2$
- "Insert identity" in discrete form: $\sum_n |n\rangle\langle n| = \hat{\mathbb{1}}$

You should be able to:
- Expand a state in a basis and compute probabilities
- Translate between $|\psi\rangle$ and its coordinate column vector in a chosen basis

Tier 3 — Advanced Undergraduate (AdvUG)

Goal: Understand why the formalism is *internally consistent* and geometric.

Core competencies:
- **Cauchy–Schwarz inequality**: $|\langle\phi|\psi\rangle|^2 \leq \langle\phi|\phi\rangle\langle\psi|\psi\rangle$
- **Triangle inequality** for norms: $\|\phi + \psi\| \leq \|\phi\| + \|\psi\|$
- Metric/geometric viewpoint: overlaps behave like cosines of angles (up to phase); distance from $\|\phi - \psi\|$ (with phase caveats)

You should be able to:
- Prove basic inequalities in Hilbert space
- Explain why overlaps and probabilities are bounded by 1 for normalised states
- See superposition as linear structure, not mystery

Tier 4 — Master's (MSc)

Goal: Shift from "vectors as states" to *rays as states* and connect to symmetry theory.

Core competencies:
- **Rays / projective Hilbert space**: physical states are equivalence classes $|\psi\rangle \sim e^{i\theta}|\psi\rangle$; pure states are points in $\mathbb{P}(\mathcal{H})$
- **Wigner theorem preview**: transformations preserving transition probabilities $|\langle\phi|\psi\rangle|^2$ are implemented by *unitary or antiunitary* operators

You should be able to:
- Explain why global phase is unphysical and why rays are the correct objects
- State what Wigner's theorem says and why it matters for symmetries

Tier 5 — PhD

Goal: Make all foundational statements *mathematically precise* and know where physics shorthand hides subtleties.

Core competencies:
- Precise definition of **Hilbert space** (inner product + completeness)
- **Separability** (existence of a countable orthonormal basis / countable dense subset) and why it is assumed in most QM
- Rays as **1D subspaces** (not just "vectors up to phase")
- Operator subtleties: Hermitian vs. self-adjoint (domains matter); generalised eigenvectors and distributional states (rigged Hilbert spaces)

You should be able to:
- Distinguish informal bra-ket manipulations from rigorous operator statements

- Discuss why separability matters for spectral decompositions
- Frame state space as a projective object and symmetry as structure on that space

1.3 What Changes Across Tiers (Without Changing the Artifact)

Instructor note: *The same assignment can be given to all tiers simultaneously. What changes is the* language*, the required* rigor*, and the* scope*. The table below illustrates this for the "phase invariance" concept.*

Tier	How "phase invariance" is engaged
HS	"Why doesn't the arrow's overall rotation matter for length?" — intuitive analogy, no algebra required
BegUG	Compute $\lvert e^{i\theta} c_n\rvert^2 = \lvert c_n\rvert^2$; verify numerically for a 2-component example
AdvUG	Prove phase invariance of all probabilities for an arbitrary orthonormal basis expansion
MSc	Interpret as well-definedness of the map $\lvert\psi\rangle \mapsto$ probabilities on rays in projective space $\mathbb{P}(\mathcal{H})$
PhD	Formalise rays as 1D subspaces; identify the $U(1)$ fiber structure of the Hopf fibration $S^1 \to S^{2n+1} \to \mathbb{CP}^n$

1.4 Worked Example: Bra-Ket Module I.2, Lecture L01

The following shows how a single lecture — **Module I.2, Lecture L01:** ***Why Dirac Notation? States as Rays & Superposition*** — is differentiated across all five tiers. This is the template for every lecture in the programme.

Lecture L01 topic: States, kets, rays, global phase, Born rule preview. Dual Sakurai (operational) and Dirac (abstract) perspectives.

Tier 1 (HS) — L01 engagement

Key question: What is a quantum state, informally?

Content focus:
- $|z|^2$ as probability; interference example with two paths
- Ket as a "direction arrow" in an abstract space
- Born rule as "probability = squared amplitude"
- Why rotating the arrow globally (multiplying by $e^{i\theta}$) does not change any measurement outcome

Signature problem: Given $A_1 = \frac{1}{\sqrt{2}}$, $A_2 = \frac{i}{\sqrt{2}}$, compute the total probability for a detector that sums both amplitudes. Verify it equals 1.

Tier 2 (BegUG) — L01 engagement

Key question: How do I write and work with quantum states?

Content focus:
- Normalisation $\langle\psi|\psi\rangle = 1$; components $c_n = \langle n|\psi\rangle$
- Notation dictionary: ket $\leftrightarrow$ column vector $\leftrightarrow$ wavefunction

- Build a 2-state example (spin-$\frac{1}{2}$) explicitly

Signature problem: Write $|\psi\rangle = \frac{3}{5}|+\rangle + \frac{4}{5}|-\rangle$. Verify normalization. Find $P(+)$ and $P(-)$. Change the global phase by $e^{i\pi/3}$ and confirm probabilities are unchanged.

Tier 3 (AdvUG) — L01 engagement

Key question: Why is global phase exactly unobservable?

Content focus:

- Cauchy–Schwarz as bound on all overlaps
- Prove that $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ give identical probabilities for *any* observable in *any* orthonormal basis
- Discuss why *relative* phase (between components) is physical

Signature problem: Prove that for any normalised $|\psi\rangle$, any orthonormal basis $\{|n\rangle\}$, and any $\theta \in \mathbb{R}$: $|\langle n|e^{i\theta}|\psi\rangle|^2 = |\langle n|\psi\rangle|^2$ for all n, and conclude the total probability is also invariant.

Tier 4 (MSc) — L01 engagement

Key question: What is the correct mathematical object for a quantum state?

Content focus:

- Rays as equivalence classes; projective Hilbert space $\mathbb{P}(\mathcal{H})$
- The Born-rule map $\mathbb{P}(\mathcal{H}) \to [0, 1]$ is well-defined on rays
- Preview: Wigner's theorem classifies symmetries of $\mathbb{P}(\mathcal{H})$

Signature problem: Define the equivalence relation on $\mathcal{H} \setminus \{0\}$ that defines rays. Show the Born-rule probability $|\langle a|\psi\rangle|^2$ depends only on the ray $[\psi]$, not the representative $|\psi\rangle$. What would it mean for a map between physical states to be a "symmetry"?

Tier 5 (PhD) — L01 engagement

Key question: What foundational structures does quantum mechanics actually require?

Content focus:

- Precise Hilbert space definition; role of completeness and separability
- Rays as 1D subspaces; $\mathbb{P}(\mathcal{H})$ as a metric space (Fubini–Study metric)
- Wigner's theorem (statement in full): every symmetry of $\mathbb{P}(\mathcal{H})$ lifts to a unitary or antiunitary operator on $\mathcal{H}$

Signature problem: (a) State the definition of a separable Hilbert space. (b) Why does separability guarantee a countable orthonormal basis? (c) State Wigner's theorem precisely, including the lifting ambiguity ($\pm$ and $U(1)$ phase). (d) Give one example of an antiunitary symmetry in QM and explain its physical significance.

2 Assessment Artifact Set

Design goals. Each lecture should assess *four distinct capabilities*:

1. **Conceptual understanding** — can you explain it?
2. **Procedural fluency** — can you compute it?
3. **Structural mastery** — can you prove or derive it?
4. **Transfer & creation** — can you apply it to a new context, or ask new questions?

The six artifact types defined in this section map cleanly onto these capabilities and are reusable across every lecture of every module in the programme.

2.1 Capability–Artifact Mapping

Capability	Artifact(s)
Conceptual understanding	Concept Questions (CQ)
Procedural fluency	Simple Problem Set (SPS)
Structural mastery	Complex Problem Set (CPS)
Transfer (engineering)	Simple Project (SPJ)
Transfer (synthesis)	Complex Project (CPJ)
Creation & frontier thinking	Well-Defined Research Question (WRQ); Open-Ended Research Question (ORQ)

2.2 Artifact 1: Concept Questions (CQ)

CQ — Concept Questions

Purpose: Fast checks that a student has the *right mental model*, not just algebra.

Deliverable format:

- 6–10 short questions: mix of multiple-choice, true/false + justify, and 2–3 sentence prompts
- Designed for 10–15 minutes in class or as pre-lecture warmup

Recommended counts by tier:

Tier	Question count & character
HS	6 (mostly qualitative)
BegUG	8 (qualitative + very short computations)
AdvUG	10 (include "explain why")
MSc / PhD	10 (include "state assumptions" + "what fails if…")

CQ Rubric

- **2 points each:** correct answer *and* explanation consistent
- Total: 12–20 points depending on count

Instructor note: *Example CQ prompts (Module I.2 Lecture L01 style):*
(1) Why is $|\psi\rangle$ not "a column vector" until a basis is chosen?
(2) What is physically meaningful: $\langle a|\psi\rangle$ or $|\langle a|\psi\rangle|^2$?
(3) Why does a global phase not change probabilities?
(4) In what sense is "superposition" basis-dependent?

2.3 Artifact 2: Simple Problem Set (SPS)

SPS — Simple Problem Set

Purpose: Build *fluency*: compute probabilities, normalise states, insert identity, perform basic basis changes.

Deliverable format:
- 8–12 problems
- Mostly 1–5 steps each
- Minimal proofs; focus on correct use of notation and technique

Suggested internal structure:
- **Part A — mechanics:** normalisation, components, overlaps
- **Part B — interpretation:** what does the computed number mean?
- **Part C — translation:** ket ↔ coordinates ↔ wavefunction preview

Tier scaling:

HS	Numeric complex arithmetic + simple amplitude-to-probability
BegUG	Normalisation + components $c_n = \langle n|\psi\rangle$ + probabilities
AdvUG	Include "show that" using completeness, but not full proofs
MSc/PhD	Fewer but deeper; emphasise representation-independence checks

SPS Rubric

Correct result	60%
Correct method/notation	30%
Interpretation sentence	10%

2.4 Artifact 3: Complex Problem Set (CPS)

CPS — Complex Problem Set

Purpose: Assess *structural mastery*: proofs, derivations, multi-step synthesis, "why the formalism works."

Deliverable format:
- 4–6 problems, multi-part (a)(b)(c)...
- Often proof-based or derivation-based
- Designed for 3–6 hours total work

Canonical CPS themes in early lectures:
- Prove Cauchy–Schwarz and/or triangle inequality

- Prove phase invariance and ray equivalence
- Show unitaries preserve probabilities and inner products
- Derive completeness consequences (Parseval identity)

Tier scaling:

AdvUG	Full proofs with guidance allowed
MSc	Proofs + conceptual interpretation + links to projective space
PhD	Proofs + additional conditions (boundedness, domains, separability)

CPS Rubric

Correct statement of assumptions	15%
Logical proof structure	45%
Correct algebra/steps	25%
Insight / interpretation	15%

2.5 Artifact 4: Simple Project (SPJ)

SPJ — Simple Project

Purpose: Hands-on transfer: "I can build something that uses the formalism."

Scope: 2–6 hours; structured, well-scaffolded; works for mixed cohorts.

Deliverable format:
- Short report (1–3 pages) *and* a small artifact: code, worksheet, or visualisation

Example SPJ Template — Basis Expansion Calculator (Module I.2, L01)

Inputs: coefficients c in basis $\{|n\rangle\}$; unitary matrix U.
Outputs: new coefficients $c' = Uc$.

Required checks:

- Normalisation preserved: $\|c'\|^2 = \|c\|^2 = 1$
- Probabilities sum to 1 in new basis
- Global phase invariance: multiplying c by $e^{i\theta}$ leaves all $|c'_n|^2$ unchanged

Deliverables: code + README; one worked example; short explanation of what changed and what did not.

SPJ Rubric

Correctness	40%
Verification tests included	25%
Clarity of explanation	25%
Reproducibility (instructions)	10%

2.6 Artifact 5: Complex Project (CPJ)

CPJ — Complex Project

Purpose: Synthesis and conceptual transfer: "I can compare frameworks and defend choices."

Scope: 8–15 hours; open-ended but guided; should produce a mini-essay or mini-tool with reflection.

Deliverable format:

- 5–8 page report + appendix with computations + optional code for basis transforms/plots

Example CPJ Template — Compare Representations (Module I.2, L01)

Choose a state and represent it in four ways:

1. Abstract ket $|\psi\rangle$ (representation-free)
2. Coordinates in basis A: $c_n = \langle n|\psi\rangle$
3. Coordinates in basis B (unitarily related to A)
4. Preview: wavefunction $\psi(x) = \langle x|\psi\rangle$ (conceptual only)

Analyse:

- What is *invariant* (norm, transition probabilities)?
- What *changes* (coordinates, basis-dependent "superposition")?

CPJ Rubric

Technical correctness	30%
Conceptual synthesis	35%
Depth & structure	25%
Originality / insight	10%

2.7 Artifact 6: Well-Defined Research Question (WRQ)

WRQ — Well-Defined Research Question

Purpose: Train "research-like thinking" while still being *bounded*: clear scope, clear output, answerable within a few days.

Scope: 5–10 hours; must be answerable with literature + calculation + interpretation.

Deliverable format — 2–6 page memo containing:

- Question statement
- Method (how you will answer it)
- Evidence (citations + computations)
- Conclusion
- 5–10 references

Example prompts (Module I.2, early bra-ket):

- "How did Born's probabilistic interpretation change the meaning of the wavefunction between 1926–1927?"
- "What exactly does 'state is a ray' buy us operationally? Provide examples where vectors fail but rays succeed."

WRQ Rubric

Clear scope and method	25%
Evidence quality	25%
Correct technical content	25%
Quality of argument	25%

2.8 Artifact 7: Open-Ended Research Question (ORQ)

ORQ — Open-Ended Research Question

Purpose: Expose the frontier: no single correct answer; evaluates *quality of reasoning*, not correctness of conclusion.

Scope: 10–20 hours (or ongoing); may include proposing a new definition, comparing frameworks, or designing a thought experiment.

Deliverable format — 4–10 page proposal-style note:

- Motivation
- What is known (brief review)
- What you propose
- How you would test or argue it
- Risks and limitations

Example prompts (Module I.2, early bra-ket):

- "What minimal operational axioms force complex Hilbert space rather than real?"
- "Can you formulate measurement and state update without projective collapse but still reproduce observed statistics?"

ORQ Rubric

Clarity + framing	25%
Depth of engagement	25%
Novelty and plausibility	25%
Awareness of limitations	25%

2.9 Exercise Packaging Standard (Multi-page Format)

Instructor note: *Every individual problem — whether in an SPS or CPS — should be packaged in the following four-layer format. This supports self-study, allows automated assembly into different deliverables, and makes solution-writing consistent.*

Four-layer exercise packet structure (one file-set per problem):

Layer	Content
1. Problem Statement	Clean, no hints; the only document students receive at the start
2. Suggestions / Hints	Scaffolded prompts; released at student request or after first attempt
3. Advanced Suggestions	Structural hints, alternative approaches, connections to other results
4. Detailed Solution	Full derivation + commentary + common pitfall warnings

File naming convention for repository automation:
`ex##_problem.tex`, `ex##_hints.tex`, `ex##_adv_hints.tex`, `ex##_solution.tex`

2.10 Recommended Per-Lecture Minimum Bundle

For each 90-minute lecture, a sustainable and scalable package is:

Artifact	Count
Concept Questions (CQ)	8
Simple Problem Set (SPS)	10 problems
Complex Problem Set (CPS)	4 problems
Simple Project (SPJ)	1
Complex Project (CPJ)	1
Well-Defined Research Q. (WRQ)	1
Open-Ended Research Q. (ORQ)	1

This yields:

- Enough material for a mixed cohort (simple through PhD tracks)
- A manageable grading load per lecture cycle
- A consistent cadence across the entire programme

2.11 Suggested Assessment Weights

Artifact type	**Weight**	**Rationale**
Concept Questions (CQ)	10%	Fast feedback; keeps students on track
Simple Problem Set (SPS)	30%	Core fluency; highest volume
Complex Problem Set (CPS)	30%	Structural mastery; differentiates tracks
Projects (SPJ + CPJ)	20%	Transfer and synthesis; project-based learning
Research Questions (WRQ + ORQ)	10%	Frontier thinking; rewards initiative
Total	**100%**	

Track-based grading policy:

- HS / BegUG → **Simple track**: CQ + SPS + SPJ only
- AdvUG / MSc / PhD → **Complex track**: all artifacts; CPS and CPJ are mandatory

Appendix — Quick Reference Card

Tier × Artifact Grid

	CQ	SPS	CPS	SPJ	CPJ	WRQ	ORQ
HS	✓	✓		✓			
BegUG	✓	✓		✓	(optional)		
AdvUG	✓	✓	✓	✓	✓	✓	
MSc	✓	✓	✓	✓	✓	✓	✓
PhD	✓	✓	✓	✓	✓	✓	✓

Artifact Duration Summary

Artifact	Format	Est. time	Primary output
CQ	6–10 questions	10–15 min	Answers + brief justifications
SPS	8–12 problems	2–4 hours	Worked solutions
CPS	4–6 problems	3–6 hours	Proofs + derivations
SPJ	Structured task	2–6 hours	Code / worksheet + 1–3 pg report
CPJ	Open task	8–15 hours	5–8 pg report + appendix
WRQ	Scoped question	5–10 hours	2–6 pg memo
ORQ	Open question	10–20 hours	4–10 pg proposal